

SYDNEY OBSERVATORY, Australia

WMO: 947680

Lat: 33.8608S

Lon: 151.2050E

Elev: 132

StdP: 14.63

Time Zone: 10.00 (AUS)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	45.0	46.4	26.9	21.0	61.0	29.4	23.6	59.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.9	89.0	67.6	84.8	68.1	81.8	68.4	73.4	81.3	72.3	79.5	71.3	78.1	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.1	115.1	76.9	70.0	110.9	76.0	69.0	106.9	75.4	37.1	81.7	36.1	79.5	35.2	78.4	78.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
N/A	N/A	N/A	DB	42.1	99.9	1.3	5.6	41.2	104.0	40.4	107.3	39.7	110.4	38.8	114.5
			WB	37.7	75.2	1.3	1.6	36.8	76.4	36.0	77.3	35.3	78.2	34.4	79.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	65.2	73.5	73.4	71.3	66.5	61.5	57.2	55.9	57.7	62.0	65.3	68.1	71.1	(6)
	DBStd	7.41	4.21	3.84	3.69	3.87	3.73	3.27	3.25	3.96	4.85	5.23	4.97	4.48	(7)
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
	HDD65	1099	0	0	2	26	121	236	283	231	118	57	21	4	(9)
	CDD50	5565	730	656	661	495	356	215	183	238	359	475	542	655	(10)
	CDD65	1190	265	236	198	71	12	0	1	4	26	68	114	195	(11)
	CDH74	5260	1321	1121	670	193	24	0	1	19	165	388	515	843	(12)
	CDH80	1166	302	199	86	29	0	0	0	2	39	132	161	216	(13)
(14) Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(14)
(15) Precipitation	PrecAvg	49.3	4.7	4.9	5.6	4.7	4.4	4.9	2.6	3.7	2.6	3.4	4.2	3.3	(15)
	PrecMax	80.2	15.2	24.8	15.3	20.0	14.6	16.1	7.2	19.0	8.9	11.2	20.4	11.0	(16)
	PrecMin	24.6	0.2	0.4	0.3	0.2	0.3	0.2	0.1	0.0	0.1	0.0	0.4	0.1	(17)
	PrecStd	13.1	3.2	4.6	3.8	4.3	3.4	3.9	2.0	4.3	2.2	3.1	4.1	2.6	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	95.2	91.3	87.1	84.0	77.3	70.3	71.9	76.5	85.0	91.0	92.8	93.1	(19)
		MCWB	68.5	72.9	69.0	64.2	60.7	58.2	56.3	56.7	60.7	62.6	65.8	66.2	(20)
	2%	DB	86.7	84.8	81.7	78.2	73.1	67.2	67.2	71.3	78.8	83.2	84.1	85.4	(21)
		MCWB	69.8	71.5	69.2	64.5	60.6	56.9	53.9	56.0	58.8	62.6	65.7	67.8	(22)
	5%	DB	82.4	81.8	79.5	75.8	70.5	65.1	64.5	68.1	73.9	77.6	79.2	81.1	(23)
		MCWB	70.7	70.8	68.5	64.4	59.9	56.2	53.4	55.1	58.0	61.9	66.0	68.1	(24)
	10%	DB	80.0	79.7	77.6	73.6	68.5	63.4	62.3	65.5	70.5	73.8	75.7	78.1	(25)
		MCWB	69.6	69.7	67.7	63.5	59.4	55.5	52.5	54.1	57.8	61.3	65.0	67.5	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	74.6	75.4	72.9	69.2	66.0	61.6	60.3	61.6	64.8	68.1	71.4	73.1	(27)
		MCDB	83.2	83.4	79.8	76.0	69.8	66.6	66.5	69.3	75.0	78.3	81.2	81.5	(28)
	2%	WB	73.0	73.8	71.6	67.8	64.2	60.0	57.6	59.3	62.9	66.1	69.3	71.3	(29)
		MCDB	81.4	81.2	77.6	74.5	69.8	64.1	62.4	66.4	71.5	75.6	77.8	79.6	(30)
	5%	WB	72.0	72.6	70.5	66.6	62.7	58.5	56.0	57.8	61.5	64.8	68.0	70.1	(31)
		MCDB	79.3	79.5	76.6	72.9	67.8	62.8	61.0	64.8	69.5	72.9	75.5	77.6	(32)
	10%	WB	71.1	71.4	69.4	65.4	61.0	57.2	54.5	56.1	60.1	63.5	66.7	69.0	(33)
		MCDB	77.8	77.9	75.6	71.5	66.6	61.8	59.9	63.2	67.8	71.0	73.6	76.0	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	10.9	10.7	11.2	13.1	13.6	12.7	14.2	15.2	15.1	13.8	12.0	11.6	(35)
		MCDBR	16.0	14.1	14.2	16.1	16.4	15.2	17.1	18.7	21.1	21.5	19.5	17.7	(36)
	5% WB	MCWBR	6.3	5.6	6.2	6.9	7.8	8.0	8.5	8.9	8.7	8.6	7.6	6.8	(37)
		MCWBR	12.9	12.6	11.8	13.0	12.7	12.1	13.7	15.2	16.1	15.5	14.9	13.9	(38)
(40) Clear-Sky Solar Irradiance	taub		0.394	0.395	0.359	0.333	0.319	0.311	0.301	0.310	0.343	0.361	0.384	0.395	(40)
		taud	2.463	2.478	2.567	2.616	2.603	2.621	2.631	2.599	2.501	2.482	2.456	2.437	(41)
	Ebn at Noon		299	292	292	283	270	265	274	287	292	299	300	300	(42)
		Edh at Noon	37	36	30	26	23	22	22	26	32	35	37	39	(43)
(44) All-Sky Solar Radiation	RadAvg		2062	1781	1513	1216	951	751	882	1168	1555	1809	1939	2108	(44)
	RadStd		147	151	129	101	65	70	70	62	84	148	172	164	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.83	+0.38	N/A	+1.33	N/A	N/A	N/A	-140	+301	+151	(46)
(47)	Regional (33 neighbors)	+0.76	N/A	N/A	N/A	N/A	N/A	-2	-155	+337	+176	(47)

Nomenclature: See separate page